omnia-sdk — GPU benchmark

End-to-end epoch timing on an NVIDIA L4, .svs via openslide against a preloaded .omnia container. Every figure below is read directly from the recorded run; nothing is restated from documentation.

Provenance

Recorded	**22 July 2026, 10:03**
GPU	**NVIDIA L4**
PyTorch	**2.11.0+cu128**
Epochs per configuration	**2**
Batch sizes swept	**64, 128, 256**
DataLoader workers swept	**0, 1, 2**
Configurations timed	**9**
Source	**omnia_benchmark_proof.json**

Every configuration measured

Batch	Workers	.svs (s)	.omnia (s)	Speedup
64	**0**	**31.43**	**7.21**	**4.36x**
64	1	36.73	12.33	2.98x
64	2	18.85	7.31	2.58x
128	0	31.51	7.30	4.32x
128	1	36.49	12.23	2.98x
128	2	18.72	7.80	2.40x
256	0	31.07	7.61	4.08x
256	1	36.54	12.53	2.92x
256	2	19.50	8.35	2.33x

Fastest configuration in green, slowest in amber. Both are the same software; only the DataLoader settings differ.

How to read this

The headline is 4.36x, and it is the least useful number here. It comes from the configuration with no DataLoader workers, where openslide decodes on the main process and is at its slowest.

Give openslide two workers and it improves substantially — from 36.73s to 18.72s — while the container barely moves, because preloaded data has no I/O left to parallelise. The advantage narrows to 2.33x.

Best configuration against best configuration is 2.60x (18.72s against 7.21s). That is the figure to plan with. Anyone quoting 4.36x is comparing their tuned setup against an untuned one.

What this run does not show

— Disk size. No container or slide size was recorded here, so the compression figures quoted elsewhere are not evidenced by this run.

— GPU utilisation. Not sampled. The claim that the format converts idle GPU time into training time is plausible from the timings but is not measured in this file.

— Data-loading share. These are whole-epoch times. The split between decoding and training was not recorded, so the often-quoted decode percentage does not come from here.

— Lossless round-trip. Verified by the test suite, not by this benchmark.

This report was produced on an NVIDIA L4 and predates omnia-sdk v1.0.0; it was recorded against the omnia-13x prototype of the same container format. It is not the Tesla T4 run quoted in the project README, and the two should not be conflated. Reproduce with: python -m omnia_sdk.benchmark

github.com/mishel-0/omnia-sdk — MIT